

4-3 LL(1)文法

1. 求FIRST集和FOLLOW集

$$S \rightarrow aAB|bA|\epsilon$$

$$A \rightarrow aAb|\epsilon$$

$$B \rightarrow bB|\epsilon$$

$$\text{FIRST}(S) = \{a, b, \epsilon\}$$

$$\text{FIRST}(A) = \{a, \epsilon\}$$

$$\text{FIRST}(B) = \{b, \epsilon\}$$

$$\text{FOLLOW}(S) = \{ \$ \}$$

$$\text{FOLLOW}(A) = \{b, \$\}$$

$$\text{FOLLOW}(B) = \{ \$ \}$$

2.

1) 不是LL(1)文法

$$S \rightarrow AB|CDA$$

$$A \rightarrow ab|c$$

$$B \rightarrow dE$$

$$C \rightarrow eC|\epsilon$$

$$D \rightarrow fd|f$$

$$E \rightarrow dE|\epsilon$$

提取公因子
→

$$S \rightarrow AB|CDA$$

$$A \rightarrow ab|c$$

$$B \rightarrow dE$$

$$C \rightarrow eC|\epsilon$$

$$D \rightarrow fd'$$

$$D' \rightarrow D|\epsilon$$

$$E \rightarrow dE|\epsilon$$

$$\text{FIRST}(S) = \{a, c, e, f, \epsilon\} \quad \text{FOLLOW}(S) = \{ \$ \}$$

$$\text{FIRST}(A) = \{a, c\} \quad \text{FOLLOW}(A) = \{d\}$$

$$\text{FIRST}(B) = \{d\} \quad \text{FOLLOW}(B) = \{ \$ \}$$

$$\text{FIRST}(C) = \{e, \epsilon\} \quad \text{FOLLOW}(C) = \{f\}$$

$$\text{FIRST}(D) = \{f\} \quad \text{FOLLOW}(D) = \{a\}$$

$$\text{FIRST}(D') = \{f, \epsilon\} \quad \text{FOLLOW}(D') = \{a\}$$

$$\text{FIRST}(E) = \{d, \epsilon\} \quad \text{FOLLOW}(E) = \{ \$ \}$$

2) 不是LL(1)文法

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